

PERIODS OF OKA K3 SURFACES ARE DENSE

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ABSTRACT. We give the first known examples of Oka Calabi–Yau manifolds in any dimension: every smooth hypersurface of multidegree $(2, \dots, 2)$ in a product of $\mathbb{P}^1$ is Oka. Building on this example, we prove that in every local Kuranishi family of K3 surfaces the points with Oka fibers form a dense G_δ subset of the base, and the sublocus of Oka fibers with vanishing Néron–Severi group is likewise a dense G_δ subset. We also obtain projective and nonprojective Oka Kummer surfaces and prove an analogous density statement in the Kummer period domain. We also prove the existence of Oka Enriques surfaces and Oka irreducible hyperkähler fourfolds.

1. INTRODUCTION

A complex manifold Y is Oka if, for every integer $n \geq 1$, every compact convex set $C \subset \mathbb{C}^n$, and every holomorphic map from a neighbourhood of C to Y , the map can be uniformly approximated on C by entire maps $\mathbb{C}^n \rightarrow Y$ [10]. This is the convex approximation property (CAP), which Forstnerič proved to be equivalent to several other classical Oka properties; see also [11, Ch. 5].

The Oka principle has its roots in the work of Oka and Grauert [27, 15], while modern Oka theory began with Gromov’s seminal paper [17]. Gromov placed the theory in the framework of the h -principle and introduced the notions of elliptic manifolds and dominating sprays. Building on Gromov’s framework, Forstnerič introduced subellipticity and proved that subelliptic manifolds are Oka [9]. His work on CAP and its variants led to the modern formulation of the Oka property [10, 11]. Despite these general criteria, deciding whether a given complex manifold is Oka remains difficult. A central problem in Oka theory is to produce new examples of Oka manifolds and to identify concrete geometric structures that imply the Oka property.

This problem is already subtle for compact complex surfaces. Forstnerič and Lárusson made a systematic study of their holomorphic flexibility properties [13]. Among the compact Oka surfaces known from their analysis are rational surfaces, ruled surfaces over $\mathbb{P}^1$ or an elliptic curve, complex tori, bielliptic surfaces, Kodaira surfaces, minimal Hopf and Enoki surfaces [13, Th. 4 and § 1]. The Oka problem for K3 surfaces had already been raised in [12, Problem C] where it is proved that every Kummer K3 surface is strongly dominable, but it left open whether any Kummer surface, or any K3 surface, is Oka [13, Cor. 3 and § 1]. More recently, Kummer surfaces

and elliptic K3 surfaces were shown to satisfy the weaker Oka-1 property, while their Oka property remained open [1, Prop. 8.4, Cor. 8.6, and § 8.2]. Prior to the present work, no example of an Oka K3 surface was known.

We fill this gap and show that Oka K3 surfaces are abundant. By a local Kuranishi family $\pi : \mathcal{X} \rightarrow B$ of a K3 surface X_0 we mean a proper holomorphic deformation with central fiber X_0 , where B is a smooth complex manifold, such that the Kodaira–Spencer map $T_0B \rightarrow H^1(X_0, T_{X_0})$ is an isomorphism. Recall that a G_δ subset of a topological space is a countable intersection of open subsets.

Theorem A. *Let $\pi : \mathcal{X} \rightarrow B$ be a local Kuranishi family of a K3 surface with fibers $X_b, b \in B$. Then both*

$$B_{\text{Oka}} = \{b \in B \mid X_b \text{ is Oka}\}, \text{ and } \{b \in B_{\text{Oka}} \mid \text{NS}(X_b) = 0\}$$

are dense G_δ subsets of B .

In other words, for every K3 surface, a Baire-generic member of its local deformation space is Oka. Here “generic” is understood in the sense of Baire category, not in the Zariski sense. Since a K3 surface with trivial Néron–Severi group is nonprojective and contains no curves, we obtain especially a dense G_δ set of nonprojective, curve-free Oka K3 surfaces in every local deformation space.

The density in Theorem A, together with the variety of examples produced below and in a forthcoming paper of ours, leads us to the following conjecture.

Conjecture. *Every K3 surface is Oka.*

For $m \geq 2$, a smooth hypersurface of multidegree $(2, \dots, 2)$ in $(\mathbb{P}^1)^{m+1}$ is a Calabi–Yau manifold, called a *Wehler Calabi–Yau m -fold* (see [7]). When $m = 2$ it is a K3 surface. The geometric starting point of the proof of Theorem A is the following explicit families of projective Oka Calabi–Yau manifolds.

Theorem B (Cor. 3.4). *Every Wehler Calabi–Yau manifold is Oka.*

By using Wehler K3 surfaces, we can also prove

Theorem C (Cor. 3.5). *Every unnodal Enriques surface is Oka.*

and

Theorem D (Thm. 6.5). *There exist irreducible hyperkähler fourfolds which are Oka.*

We note that in any moduli space of polarized Enriques surfaces, the unnodal ones form a dense Zariski-open locus. We don’t know whether any or every nodal Enriques surface is Oka or not.

Theorem A can also be formulated in Hodge-theoretic terms. For background on markings, the K3 lattice, and the period domain, see [19]. A

marking of a K3 surface X is an isometry $\varphi : H^2(X, \mathbb{Z}) \rightarrow \Lambda$, where $\Lambda = 3U \oplus 2E_8(-1)$ is the K3 lattice, U has Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and $E_8(-1)$ is the negative definite E_8 lattice. If $\omega \in H^{2,0}(X)$ is a non-zero 2-form, then the *period* is defined to be the oriented positive plane $P_X = \varphi(\text{span}_{\mathbb{R}}\{\text{Re } \omega, \text{Im } \omega\})$. The *K3 period domain* $\mathcal{D}$ is defined to be the space of oriented two-planes on which the form of $\Lambda_{\mathbb{R}}$ is positive definite. We denote by $\text{NS}(X) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X)$ the Néron–Severi lattice. For a marking φ , we have $\varphi(\text{NS}(X)) = \Lambda \cap P_X^{\perp}$. For a complex two-torus A , we denote by $\text{Kum}(A)$ the minimal resolution of $A/\{\pm 1\}$. Such a surface is K3 and is called a *Kummer surface*. Let $A[2]$ be the subgroup of 2-torsion points. The quotient $A/\{\pm 1\}$ has sixteen nodes. If E_a , $a \in A[2]$, are the exceptional curves (of the resolution) on $\text{Kum}(A)$, then the lattice

$$\text{Kum} := H^2(\text{Kum}(A), \mathbb{Z}) \cap \text{span}_{\mathbb{Q}}\{[E_a] \mid a \in A[2]\}$$

is called the *Kummer lattice*. It is negative definite of rank 16 and has discriminant 2^6 . Its primitive embedding in Λ is unique up to composition with elements of the orthogonal group $O(\Lambda)$ [19, Ch. 14, Cor. 3.15]. Fix such an embedding $\text{Kum} \subset \Lambda$, let $L := \text{Kum}^{\perp}$, and let $\mathcal{D}_K := \{P \in \mathcal{D} \mid P \perp \text{Kum}\}$. Then L has signature $(3, 3)$, $\dim_{\mathbb{C}} \mathcal{D} = 20$, and $\dim_{\mathbb{C}} \mathcal{D}_K = 4$. Nikulin’s criterion [26, Th. 3] says that a K3 surface is Kummer if and only if its Néron–Severi lattice contains a primitive copy of Kum . Hence every period in $\mathcal{D}_K$ represents a marking of a Kummer surface.

Theorem E. *Let $\mathcal{O} \subset \mathcal{D}$ be the set of periods represented by marked K3 surfaces which are Oka, and let $\mathcal{O}_K := \mathcal{O} \cap \mathcal{D}_K$.*

- (1) *The set $\mathcal{O}$ is dense G_{δ} in $\mathcal{D}$, and so is $\{P \in \mathcal{O} \mid \Lambda \cap P^{\perp} = 0\}$.*
- (2) *The set $\mathcal{O}_K$ is dense G_{δ} in $\mathcal{D}_K$, and so is $\{P \in \mathcal{O}_K \mid \Lambda \cap P^{\perp} = \text{Kum}\}$.*

In particular, there exist projective and nonprojective Oka Kummer surfaces.

The construction leading to the theorem also produces a positive-dimensional family of projective Oka Kummer surfaces. In a forthcoming paper, we will prove that every projective Kummer surface is Oka. We note that the density of Oka parameters asserted by Theorem E works on the period domain which is a Hodge-theoretic parameter space, and it does not imply density of Oka parameters in a moduli of polarized K3 surfaces.

2. STRATEGY AND PLAN

In Section 3 we prove Theorem B. The idea is as follows. Whenever a Calabi–Yau manifold admits a one-dimensional fibration, or whenever a hyperkähler manifold admits a Lagrangian fibration, one can use the holomorphic volume form or the holomorphic symplectic form to build a holomorphic spray. A Wehler Calabi–Yau manifold admits naturally fibrations onto products of $\mathbb{P}^1$ for which the corresponding one-dimensional sprays form a dominating family. This construction could also work for some other K3 surfaces with two elliptic fibrations, especially those with rich dynamics, see

Cor. 3.8. The method is flexible and could also be adapted to some other situations. Theorem C about Oka Enriques surfaces is a quick consequence. However the construction of the hyperkähler example is technically more involved and is thus postponed to Section 6, though the mechanism of the construction is the same as those in Section 3.

In Section 4 we discuss one very special example of K3 surface: the Fermat quartic. Drawing on classical algebraic geometry of pencils of hyperplane sections, we show that it is simultaneously a K3 surface of type $(2, 2, 2)$ and a Kummer surface. Therefore Theorem B implies the existence of an Oka Kummer surface. Such classical algebro-geometric arguments also play a crucial role in the construction of open Oka varieties by the first author [33]. We will also show that the particular needed feature of the Fermat quartic survives in a positive-dimensional deformation family. This deformation is crucial for our later application.

In Section 5 we apply Ratner's theory to obtain the density assertion in Theorem A, while the G_δ property is already available by Lárússon's work [22, Cor. 8]. Ratner [29] established deep classification theorems for orbit closures of unipotent flows on homogeneous spaces. Verbitsky [31, 32] applied this theory to the action of the K3 modular group on the period domain. In particular Verbitsky asserts that a totally irrational K3 period has dense orbit in the period domain. Our deformation of the Fermat quartic constructed in Section 4 provides Oka marked K3 surfaces with such totally irrational periods.

3. HOLOMORPHIC FORMS, FIBRATIONS AND SPRAYS

Gromov [17] introduced the notion of spray. A *spray* on a complex manifold X is a holomorphic vector bundle $p : E \rightarrow X$ and a holomorphic map $s : E \rightarrow X$ such that $s(0_x) = x$. Its vertical differential at 0_x is the restriction of ds_{0_x} to the fiber direction E_x . A finite set of sprays is dominating if the images of these vertical differentials span $T_x X$ at every point. A manifold carrying a dominating family is called *subelliptic*. It is known that every subelliptic manifold is Oka (see [9] and [11, Cor. 5.6.14]). We say that a spray $s : E \rightarrow X$ with bundle projection $p : E \rightarrow X$ *preserves* a fibration $f : X \rightarrow B$ if $f \circ s = f \circ p$. In this case $s(E_x)$ is contained in the fiber $f^{-1}(f(x))$ for every x .

3.1. Vertical integration. Let $f : X \rightarrow B$ be a holomorphic map between complex manifolds and let $\pi : E \rightarrow B$ be a holomorphic vector bundle. A point of f^*E is a pair (x, λ) with $\lambda \in E_{f(x)}$.

We will say that a spray $s : f^*E \rightarrow X$ is *additive over B* if it preserves f and if

$$(3.1) \quad s(s(x, \lambda), \mu) = s(x, \lambda + \mu), \quad \lambda, \mu \in E_{f(x)}.$$

For a local section a of an additive E , we define

$$(3.2) \quad \tau_a(x) := s(x, a(f(x))).$$

By (3.1), this is a local biholomorphism with inverse τ_{-a} .

Assume from now on that $A : f^*E \rightarrow T_X$ is a holomorphic morphism of vector bundles such that $df \circ A = 0$. For a local section a of E , we define its *vertical vector field* by $V_a(x) := A_x(a(f(x)))$.

Lemma 3.1. *There is a spray on f^*E which preserves f and has vertical differential A . It is additive if $[V_a, V_b] = 0$ for all local sections a, b defined on a common open set. This condition always holds when E has rank one.*

Proof. We consider f^*E as a complex submanifold of $X \times E$. A vector $(v, \eta) \in T_x X \times T_\lambda E$ is tangent to f^*E if and only if $df_x(v) = d\pi_\lambda(\eta)$. Define locally a vector field on f^*E by $\tilde{V}_{(x, \lambda)} := (A_x(\lambda), 0)$, where 0 is the zero vector in $T_\lambda E$. Since $df \circ A = 0$, the local definitions glue to a globally defined holomorphic vector field $\tilde{V}$ on f^*E .

The flow of $\tilde{V}$ through (x, λ) remains in the compact set $f^{-1}(f(x)) \times \{\lambda\}$ and exists for every complex time. Its time-one map, composed with the projection onto X , gives rise to a spray s preserving f . At $\lambda = 0$ the vector field vanishes, and its first variation in the parameter is A_x . Thus the vertical differential of the spray s is A .

If the V_a commute, then the flow of a sum is the composition of their flows, and we obtain (3.1). In the rank-one case, two such vector fields have the form $(u \circ f)V$ and $(v \circ f)V$, where u, v are holomorphic functions on the base and V is vertical. Since V annihilates both coefficients, their bracket vanishes. $\square$

3.2. Holomorphic symplectic forms. A *holomorphic symplectic form* is a closed, everywhere nondegenerate holomorphic 2-form. Now we consider a holomorphic symplectic manifold X equipped with a such a form σ .

For a tangent vector v and a differential k -form η , we denote by $\iota_v \eta$ the contraction of η with v , i.e. $(\iota_v \eta)(v_1, \dots, v_{k-1}) = \eta(v, v_1, \dots, v_{k-1})$. We denote the inverse of the contraction $v \mapsto \iota_v \sigma$ by $\sigma^\sharp : T_X^* \rightarrow T_X$. For a subspace $W \subset T_x X$, its *symplectic orthogonal* is defined as $W^\perp = \{v \in T_x X \mid \sigma_x(v, w) = 0 \text{ for all } w \in W\}$. A smooth submanifold $Y \subset X$ is called *coisotropic* if $(T_x Y)^\perp \subset T_x Y$ at every $x \in Y$, and *Lagrangian* if equality holds. In particular, smooth hypersurfaces are coisotropic. A fibration on X is called *Lagrangian* if its smooth fibers are Lagrangian.

If $f : X \rightarrow B$ is a proper surjective holomorphic map between complex manifolds, and if the smooth fibers of f are coisotropic, then we can define a morphism of bundles $A_f : f^*T_B^* \rightarrow T_X$ by

$$(3.3) \quad A_f := \sigma^\sharp \circ df^*, \quad \sigma_x(A_{f,x}(\lambda), v) = \lambda(df_x(v)).$$

Let $f_j : X \rightarrow B_j$, $1 \leq j \leq k$, be finitely many maps satisfying the above hypotheses.

Proposition 3.2. *The morphism A_f is the vertical differential of an additive spray on T_B^* . If X is compact and $(f_1, \dots, f_k)$ is an immersion, then the corresponding sprays form a dominating family, hence X is Oka.*

Proof. Over the regular values of f , we have $\text{im } A_f = (\ker df)^\perp \subset \ker df$. The regular locus being dense and open, we have $df \circ A_f = 0$ everywhere.

Let a, b be holomorphic 1-forms on an open set $U \subset B$. On $f^{-1}(U)$, recall that we have vector fields $V_a(x) = A_{f,x}(a(f(x)))$ and $V_b(x) = A_{f,x}(b(f(x)))$. By (3.3), we have $\iota_{V_a}\sigma = f^*a$ and $\iota_{V_b}\sigma = f^*b$. Since $df(V_a) = df(V_b) = 0$, for every holomorphic differential form η of positive degree on U we have $\iota_{V_a}(f^*\eta) = \iota_{V_b}(f^*\eta) = 0$.

We denote by $\mathcal{L}_V$ the Lie derivative along a vector field V . By Cartan's formula and $d\sigma = 0$, we obtain $\mathcal{L}_{V_a}\sigma = d(\iota_{V_a}\sigma) = f^*(da)$. Moreover, $\mathcal{L}_{V_a}(f^*b) = d(\iota_{V_a}(f^*b)) + \iota_{V_a}(f^*(db)) = 0$. We get thus

$$\iota_{[V_a, V_b]}\sigma = \mathcal{L}_{V_a}(\iota_{V_b}\sigma) - \iota_{V_b}(\mathcal{L}_{V_a}\sigma) = -\iota_{V_b}(f^*(da)) = 0.$$

As σ is nondegenerate, we have $[V_a, V_b] = 0$. By Lemma 3.1, we obtain the desired additive spray.

If the product map is an immersion, its dual differential is surjective. Thus the images of the df_j^* span T_x^*X . By (3.3), the vertical differentials span T_xX at every point. In other words these sprays form a finite dominating family. Hence X is subelliptic and therefore Oka by Forstnerič's theorem [9]. $\square$

3.3. Holomorphic volume forms. Let now X have dimension $m \geq 2$ and admit a nowhere-vanishing holomorphic m -form Ω , i.e. a holomorphic volume form. For a proper holomorphic map $f : X \rightarrow B$ to a smooth $(m-1)$ -fold, define a morphism of bundles $A_f : f^*K_B \rightarrow T_X$, where K_B denotes the canonical bundle of B , by

$$(3.4) \quad \iota_{A_{f,x}(\lambda)}\Omega_x := (\wedge^{m-1}(df_x)^*)(\lambda).$$

For a holomorphic map $\Phi : X \rightarrow B_1 \times \cdots \times B_N$ to a product of smooth curves and a subset $J \subset \{1, \dots, N\}$ of cardinality $m-1$, we denote by $f_J : X \rightarrow \prod_{j \in J} B_j$ the composition of Φ with the coordinate projection.

Proposition 3.3. *The morphism A_f is the vertical differential of an additive spray on f^*K_B over X . If X is compact and Φ is an immersion, then the sprays associated with the f_J form a finite dominating family, hence X is Oka.*

Proof. For $\xi \in T_{f(x)}^*B$, we obtain from (3.4) that $f^*\xi \wedge \iota_{A_f(\lambda)}\Omega = f^*(\xi \wedge \lambda) = 0$. Since the left-hand side is $\xi(df(A_f(\lambda)))\Omega$, we deduce that $df \circ A_f = 0$. Thus we can construct a spray preserving f by Lem. 3.1. The additive property of the spray follows from the rank-one case of Lem. 3.1.

If X is compact, every f_J is proper. If Φ is also an immersion, the images of the cotangent lines of the factors span T_x^*X . Their $(m-1)$ -fold exterior products therefore span $\wedge^{m-1}T_x^*X$. By (3.4), the vertical differentials span T_xX . As in the proof of Prop. 3.2 we conclude by Forstnerič's theorem [9]. $\square$

Recall that, for $m \geq 2$, a *Wehler Calabi–Yau m -fold* is a smooth hypersurface of multidegree $(2, \dots, 2)$ in $(\mathbb{P}^1)^{m+1}$; the fact that it is Calabi–Yau can be seen easily by the adjunction formula (see [7]).

Corollary 3.4. *Every Wehler Calabi–Yau manifold is Oka.*

Proof. A Calabi–Yau manifold admits a holomorphic volume form. The inclusion of a Wehler manifold into $(\mathbb{P}^1)^{m+1}$ satisfies the hypotheses of Prop. 3.3. Remark that when $m = 2$ the volume form is a symplectic form and we can also apply Prop. 3.2 to the three projections onto $\mathbb{P}^1$. $\square$

We say that a complex Enriques surface is *unnodal* if it contains no (-2) -curves.

Corollary 3.5. *Every unnodal Enriques surface is Oka.*

Proof. By [4, Th. 5.1], every unnodal Enriques surface Y has an étale double cover X which is a Wehler K3 surface. By Cor. 3.4 X is Oka. We conclude because the Oka property survives under topological coverings, see [11, Prop. 5.6.3]. $\square$

3.4. A dynamical criterion. Let X be a complex manifold. For each index i , let $f_i : X \rightarrow B_i$ be a proper holomorphic map, let $E_i \rightarrow B_i$ be a holomorphic vector bundle, and let $s_i : f_i^* E_i \rightarrow X$ be an additive spray, with vertical differential A_i . Define the set

$$(3.5) \quad U := \left\{ x \in X \mid \sum_i \operatorname{im}(A_{i,x}) = T_x X \right\}.$$

The set U is open because the spanning condition at one point involves only finitely many vectors which persist in a neighborhood of that point.

For a holomorphic section a of E_i over an open subset of B_i , Recall that we have defined in (3.2) a local biholomorphism $\tau_{i,a}(x) = s_i(x, a(f_i(x)))$. Let $\mathcal{P}$ be a pseudogroup generated by such local biholomorphisms and the group $\operatorname{Aut}(X)$. The $\mathcal{P}$ -orbit of a point x is the set of points obtainable from x by finite compositions of the generators and their inverses, wherever these compositions are defined.

Following [21], for a holomorphic map $h : S \rightarrow X$, we will say that a *spray over h* with parameter space $\mathbb{C}^n$ is a holomorphic map $H : S \times \mathbb{C}^n \rightarrow X$ satisfying $H(s, 0) = h(s)$ for every $s \in S$. We call $d_t H(s, 0) : \mathbb{C}^n \rightarrow T_{h(s)} X$ its parameter differential.

Proposition 3.6. *If every $\mathcal{P}$ -orbit intersects U , then X is Oka.*

Proof. By a theorem of Kusakabe [21, Cor. 4.1], it suffices to prove that, for every holomorphic map $h : S \rightarrow X$ from a Stein manifold S and every $s_0 \in S$, there are finitely many sprays over h whose parameter differentials at s_0 span $T_{h(s_0)} X$.

For each i , by Cartan's theorem A [8] we can find finitely many global sections $p_{i,j}, j = 1, \dots, \nu_i$ of $(f_i \circ h)^* E_i$ spanning its fiber at s_0 . The map

$$K_i(s, t) := s_i\left(h(s), \sum_j t_j p_{i,j}(s)\right), \quad s \in S, t \in \mathbb{C}^{\nu_i}$$

defines a spray over h and satisfies $\text{im}(d_t K_i(s_0, 0)) = \text{im}(A_{i,h(s_0)})$. If $h(s_0) \in U$, then by definition of U we deduce that finitely many of these sprays over h satisfy the above Kusakabe's conditions.

We prove the assertion simultaneously for all S, h, s_0 by induction on the length of a composition of generators carrying $h(s_0)$ into U . The case of length zero is proved. If the first generator is an automorphism g , then we can apply induction to $g \circ h$ and compose the corresponding sprays by g^{-1} . Assume now that $\gamma_l \circ \dots \circ \gamma_1(h(s_0)) \in U$, and that γ_1 is a local biholomorphism $\tau_{i,a}$ where a is a local section of $E_i \rightarrow B$. By Cartan's theorem A, there exists a global section p of $(f_i \circ h)^* E_i$ with

$$(3.6) \quad p(s_0) = a(f_i(h(s_0))).$$

Then we define $h_1(s) := s_i(h(s), p(s))$. We have $h_1(s_0) = \gamma_1(h(s_0))$, so $\gamma_l \circ \dots \circ \gamma_2$ sends $h_1(s_0)$ into U . By induction, there are finitely many sprays $H_k : S \times \mathbb{C}^{n_k} \rightarrow X$ over h_1 whose parameter differentials at s_0 span $T_{h_1(s_0)} X$.

Since $f_i \circ h_1 = f_i \circ h$, the restriction of $(f_i \circ H_k)^* E_i$ to $S \times \{0\}$ coincides with $(f_i \circ h)^* E_i$. By Cartan's theorem B [8] we can extend $(-p)$ to a global section q_k of $(f_i \circ H_k)^* E_i$ on $S \times \mathbb{C}^{n_k}$. Define $\hat{H}_k : S \times \mathbb{C}^{n_k} \rightarrow X$ by

$$(3.7) \quad \hat{H}_k(s, t) = s_i(H_k(s, t), e), \text{ where } q_k(s, t) = ((s, t), e), e \in (E_i)_{f_i(H_k(s, t))}.$$

If $p(s) = (s, e)$, then $h_1(s) = s_i(h(s), e)$ and $q_k(s, 0) = ((s, 0), -e)$. By (3.1) and (3.7), we obtain $\hat{H}_k(s, 0) = s_i(h_1(s), -e) = s_i(s_i(h(s), e), -e) = h(s)$. Hence $\hat{H}_k$ is a spray over h .

Let $\pi : f_i^* E_i \rightarrow X$ be the projection. Define $\tilde{H}_k : S \times \mathbb{C}^{n_k} \rightarrow f_i^* E_i$ by $\tilde{H}_k(s, t) = (H_k(s, t), e)$, where $q_k(s, t) = ((s, t), e)$. Then we have $\pi \circ \tilde{H}_k = H_k$ and, by (3.7), $s_i \circ \tilde{H}_k = \hat{H}_k$.

Let $\xi = (h_1(s_0), -a(f_i(h_1(s_0)))) \in f_i^* E_i$. Since $q_k|_{S \times \{0\}} = -p$ and $f_i \circ h_1 = f_i \circ h$, by (3.6) we obtain that $\tilde{H}_k(s_0, 0) = \xi$ for every k . By induction, the images of $d_t H_k(s_0, 0)$ span $T_{h_1(s_0)} X$. As $d\pi|_{\xi} \circ d_t \tilde{H}_k(s_0, 0) = d_t H_k(s_0, 0)$, we deduce that

$$(3.8) \quad \ker(d\pi|_{\xi}) + \sum_k \text{im}(d_t \tilde{H}_k(s_0, 0)) = T_{\xi}(f_i^* E_i).$$

For x near $h_1(s_0)$, we have $s_i(x, -a(f_i(x))) = \tau_{i,-a}(x)$. Since $\tau_{i,-a}$ is a local biholomorphism, by the chain rule we deduce that $ds_i|_{\xi}$ is surjective. Moreover, by (3.1) and (3.6), for every $u \in (E_i)_{f_i(h(s_0))}$ we have $s_i(h_1(s_0), -a(f_i(h_1(s_0)))) + u = s_i(h(s_0), u)$. As $\ker(d\pi|_{\xi}) \simeq (E_i)_{f_i(h(s_0))}$, the restriction of $ds_i|_{\xi}$ to $\ker(d\pi|_{\xi})$ is just $A_{i,h(s_0)}$. Applying $ds_i|_{\xi}$ to (3.8),

we therefore obtain

$$(3.9) \quad \text{im}(A_{i,h(s_0)}) + \sum_k \text{im}(d_t \hat{H}_k(s_0, 0)) = T_{h(s_0)}X.$$

Since $\text{im}(d_t K_i(s_0, 0)) = \text{im}(A_{i,h(s_0)})$, by (3.9) we conclude that the parameter differentials of K_i and the $\hat{H}_k$ span $T_{h(s_0)}X$. This completes the induction, and hence X is Oka. $\square$

Corollary 3.7. *Let $\pi : X \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ be the double cover branched over a smooth curve of bidegree $(4, 4)$. The surface X is an Oka K3 surface.*

Proof. The double cover of $\mathbb{P}^1 \times \mathbb{P}^1$ branched over a smooth curve of bidegree $(4, 4)$ is a K3 surface, by a quick application of the adjunction formula (see [30, Th. 2.2]). We will fix a holomorphic symplectic form on X .

Let $f_i = \text{pr}_i \circ \pi$, where $\text{pr}_i : \mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is the i -th projection. By Prop. 3.2 we obtain additive sprays $s_i : f_i^* T_{\mathbb{P}^1}^* \rightarrow X$ with vertical differentials A_i . Let $\mathcal{P}$ be the pseudogroup defined above with respect to these sprays.

By (3.3), the images of $A_{1,x}$ and $A_{2,x}$ span $T_x X$ exactly when $d\pi_x$ is invertible. Let $R \subset X$ be the ramification curve of π . Thus the set U defined in (3.5) is $X \setminus R$.

Fix $x \in R$. As R is smooth, in suitable local coordinates centered at x and $\pi(x)$, we have $\pi(z, w) = (z, w^2)$ and $R = \{w = 0\}$. Hence $\ker(d\pi_x) \cap T_x R = \{0\}$. Let $i \in \{1, 2\}$ such that $df_{i,x} \neq 0$. Remark that $\text{im}(A_{i,x}) = \ker(df_{i,x}) = \ker(d\pi_x)$. For a local holomorphic one-form a on $\mathbb{P}^1$ with $a(f_i(x)) \neq 0$, we have

$$\left. \frac{d}{dt} \right|_{t=0} \tau_{i,ta}(x) = A_{i,x}(a(f_i(x))) \notin T_x R.$$

As a consequence, $\tau_{i,ta}(x) \in U$ for every sufficiently small $t \neq 0$. Thus every $\mathcal{P}$ -orbit meets U , and we conclude by Prop. 3.6. $\square$

Let X be a projective K3 surface. We say that $\text{Aut}(X)$ is *non-elementary* if its action on $\text{GL}(H^2(X, \mathbb{Z}))$ contains a nonabelian free subgroup. An automorphism $g \in \text{Aut}(X)$ is called *parabolic* if it has infinite order and preserves a genus-one fibration.

Corollary 3.8. *Assume that X is a K3 surface such that $\text{Aut}(X)$ is non-elementary, contains a parabolic automorphism, and preserves no nonempty proper algebraic subset of X . Then X is Oka.*

Proof. As $\text{Aut}(X)$ is non-elementary, there are two parabolic automorphisms whose invariant genus-one fibrations $f_1, f_2 : X \rightarrow \mathbb{P}^1$ are distinct (see [5]). We will fix a holomorphic symplectic form σ on X . By Prop. 3.2, we have additive sprays $s_i : f_i^* T_{\mathbb{P}^1}^* \rightarrow X$ with vertical differentials A_i . Let $\mathcal{P}$ be the pseudogroup considered above generated by $\text{Aut}(X)$ and the local biholomorphisms defined by the sprays.

The set U defined in (3.5) satisfies

$$(3.10) \quad U = \{x \in X \mid \ker(df_{1,x}) \cap \ker(df_{2,x}) = \{0\}\}.$$

As the morphism $(f_1, f_2) : X \rightarrow \mathbb{P}^1 \times \mathbb{P}^1$ is generically finite, U is a nonempty Zariski-open subset of X .

For every $x \in X$, the Zariski closure of the orbit $\text{Aut}(X) \cdot x$ is a nonempty $\text{Aut}(X)$ -invariant algebraic subset. By hypothesis it is X , so the orbit intersects U . We conclude by Prop. 3.6. $\square$

Remark 3.9. Cor. 3.8 is one of the major inspiration/motivation for our work. We refer to [6] for examples of K3 surfaces satisfying the hypothesis of Cor. 3.8.

4. THE FERMAT QUARTIC AND KUMMER SURFACES

In this section we will show that many Kummer surfaces are Oka, including some examples with totally irrational period. In the first subsection we study the Fermat quartic and in the second subsection we construct one deformation family of it.

4.1. A $(2, 2, 2)$ model. Choose a root of unity $\zeta \in \mathbb{C}$ with $i = \zeta^2$. Consider the Fermat quartic in $\mathbb{P}^3$ defined in homogeneous coordinates by $x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0$. It is a K3 surface, see [19, § 1.1.3]. Via the linear change of coordinates $x_0 = \zeta X$, $x_1 = Y$, $x_2 = \zeta Z$, $x_3 = W$, it is isomorphic to

$$Q = \{Y^4 + W^4 = X^4 + Z^4\} \subset \mathbb{P}^3.$$

It contains the three pairwise disjoint lines

$$L_1 = \{X=Y, Z=W\}, L_2 = \{X=iY, Z=-W\}, L_3 = \{X=-Y, Z=iW\}.$$

Let H denote the hyperplane class on Q . For each j , every plane through L_j cuts Q in a divisor of the form $L_j + C$, where C is a plane cubic. As the plane varies, these cubics form a pencil of divisors of class $H - L_j$ where H denotes the hyperplane class.

Lemma 4.1. *For each $j = 1, 2, 3$, the cubics cut out on Q by the planes through L_j form a base-point-free pencil whose general member is a smooth genus-one curve.*

Proof. Fix j and let l_0, l_1 be two independent linear equations defining L_j . Since $L_j \subset Q$, the defining quartic equation of Q can be written as $l_0 A + l_1 B = 0$ where A and B are some cubic forms.

For $[a : b] \in \mathbb{P}^1$, let $P_{[a:b]}$ be the plane $al_0 + bl_1 = 0$. Its intersection with Q is the union of L_j and the residual cubic

$$C_{[a:b]} = \{al_0 + bl_1 = 0, bA - aB = 0\}.$$

It is clear that the pencil has no base point outside L_j . If $p \in L_j$ were a base point, then $p \in C_{[a:b]}$ for every $[a : b] \in \mathbb{P}^1$, so $bA(p) - aB(p) = 0$ for every $[a : b]$. Consequently $A(p) = B(p) = 0$. Since $l_0(p) = l_1(p) = 0$, we would then have

$$d(l_0 A + l_1 B)_p = A(p) dl_{0p} + B(p) dl_{1p} = 0,$$

contrary to the smoothness of Q .

Thus the pencil is base-point-free. By Bertini's theorem, its general member is smooth. Each member is a plane cubic, so every smooth member has genus one. $\square$

For $j = 1, 2, 3$, let $\pi_j : Q \rightarrow \mathbb{P}^1$ be the morphism defined by the pencil of Lemma 4.1, and let F_j denote its fiber divisor class. Since a plane through L_j cuts Q in L_j and a fiber of π_j , we have the equality $F_j = H - L_j$ between divisor classes on Q . As the intersection numbers satisfy $H^2 = 4$, $HL_j = 1$, and as the lines L_j are pairwise disjoint, we obtain $F_j F_k = 2$ for $j \neq k$.

Proposition 4.2. *The morphism $\Phi : Q \rightarrow (\mathbb{P}^1)^3$ induced by the three pencils constructed above is an isomorphism onto a smooth hypersurface of multidegree $(2, 2, 2)$.*

Proof. We first prove that Φ is a finite morphism. If an irreducible curve $C \subset Q$ were contracted by all three π_j , then for each j it would lie in a plane through L_j . As the first two such planes intersect in one line, the curve C would itself be a line on Q . Remark that by construction this line would meet each L_j because it shares a plane with each of them. A direct computation with the explicit equations of the three lines L_j shows that no such line would exist.

We next prove that Φ is birational onto its image, i.e. has degree one. We prove this by showing that a general value has only one preimage. At $p = [1 : 0 : 0 : 1] \in Q$, the three planes corresponding to the fibers of π_1, π_2, π_3 through p have equations

$$X - Y + Z - W = 0, \quad X - iY - Z - W = 0, \quad X + Y - iZ - W = 0.$$

The determinant of their coefficients in X, Y, Z is 2, so these three planes intersect in the single point p . The same is true for triples of fiber planes parametrized by a nonempty Zariski-open subset of the image of Φ . In other words a general fiber of Φ consists of one point.

Denote $Y = \Phi(Q)$. Since Φ is a finite morphism, Y is an integral surface in $(\mathbb{P}^1)^3$. As $(\mathbb{P}^1)^3$ is smooth, the Weil divisor Y is Cartier (see [18, Prop. 2.6.11]). Let H_j denote the class on $(\mathbb{P}^1)^3$ of the pullback line bundle of $\mathcal{O}_{\mathbb{P}^1}(1)$ from the j -th factor of $(\mathbb{P}^1)^3$. Then $[Y] = aH_1 + bH_2 + cH_3$ where (a, b, c) is the multidegree of Y . Since $\Phi^*H_j = F_j$ and $\Phi_*[Q] = [Y]$, we obtain the intersection numbers $a = F_2F_3$, $b = F_1F_3$, and $c = F_1F_2$ by the projection formula (see [14, Prop. 2.3(c)]). Hence $a = b = c = 2$.

As Y is a Cartier divisor in $(\mathbb{P}^1)^3$, it is Cohen—Macaulay, i.e. locally complete intersection (see [18, Th. II.8.21, Prop. II.8.23]). By the adjunction formula (see [18, Th. 7.11]), the dualizing sheaf of Y is trivial. Since the dualizing sheaf of the smooth surface Q is simply its canonical bundle, it is also trivial. Hence the conductor of the finite morphism $\Phi : Q \rightarrow Y$ is trivial by [28, Prop. 2.9] (see also [18, Ex. III.7.2]). This means that Φ is an isomorphism. $\square$

Corollary 4.3. *The Fermat quartic is an Oka Kummer surface.*

Proof. By Proposition 4.2 and Corollary 3.4, the Fermat quartic is Oka. It is a Kummer surface by Mizukami's theorem, see [20, App. Th. A.1]. $\square$

4.2. Deformations preserving the three pencils. Recall that Λ is the K3 lattice from the introduction. Again we refer to [19] for background on lattices and cohomological aspects of K3 surfaces. In this subsection we will fix a marking $H^2(Q, \mathbb{Z}) \rightarrow \Lambda$ which sends its Kummer sublattice to the fixed primitive sublattice $\text{Kum} \subset \Lambda$. Via this marking, we also regard the fiber classes F_1, F_2, F_3 as elements of Λ . Let $\Lambda_Q \subset \Lambda$ be the smallest primitive sublattice containing $\text{Kum}, F_1, F_2, F_3$.

By Prop. 4.2, the class $F_1 + F_2 + F_3$ is the pullback of $\mathcal{O}_{(\mathbb{P}^1)^3}(1, 1, 1)$, thus ample. The Hodge index theorem then implies that Λ_Q is nondegenerate of signature $(1, \text{rk } \Lambda_Q - 1)$. Since $\text{rk } \text{Kum} = 16$, we have $\text{rk } \Lambda_Q \leq 19$. Hence the period domain

$$(4.1) \quad \mathcal{D}_Q = \{P \in \mathcal{D} \mid P \perp \Lambda_Q\}$$

is contained in the Kummer period domain $\mathcal{D}_K$ and has complex dimension

$$(4.2) \quad 20 - \text{rk } \Lambda_Q \geq 1.$$

The following proposition performs a deformation of Prop. 4.2. It is the key geometric input for the dynamical machinery that we investigate later in Section 5.

Proposition 4.4. *A nonempty open subset of $\mathcal{D}_Q$ parametrizes Kummer surfaces which are smooth hypersurfaces of multidegree $(2, 2, 2)$ in $(\mathbb{P}^1)^3$. All these surfaces are Oka.*

Proof. Let $p : \mathcal{X} \rightarrow T$ be a local Kuranishi family of Q over a simply connected base T , with central point $0 \in T$. Our fixed marking $H^2(Q, \mathbb{Z}) \rightarrow \Lambda$ extends to a trivialization of $R^2p_*\mathbb{Z}$, i.e. it induces markings on nearby fibers. The induced local period map $\text{per} : T \rightarrow \mathcal{D}$ is a local biholomorphism by the local Torelli theorem (see [19, Ch. 6, 7]).

Let $B := \text{per}^{-1}(\mathcal{D}_Q)$. Up to shrinking T , we can and will assume that the period map sends B biholomorphically to a neighborhood of the period of Q in $\mathcal{D}_Q$. By definition (4.1) of $\mathcal{D}_Q$, every class in Λ_Q , and in particular each F_i , is orthogonal to the period, thus of type $(1, 1)$ on every fiber over B . Therefore they are classes of line bundles. The line bundles $\mathcal{O}_Q(F_i)$ thus extend to line bundles $\mathcal{L}_i$ on $\mathcal{X}$ (see [19, Ch. 6, §2.4] for the arguments in this paragraph).

We have $\mathcal{O}_Q(F_i) \simeq \pi_i^* \mathcal{O}_{\mathbb{P}^1}(1)$ for the fibration $\pi_i : Q \rightarrow \mathbb{P}^1$, so $h^0(Q, \mathcal{O}_Q(F_i)) = 2$. Moreover, $h^2(Q, \mathcal{O}_Q(F_i)) = h^0(Q, \mathcal{O}_Q(-F_i)) = 0$ by Serre duality (see [2, § I.5]). Since the self-intersection $F_i^2 = 0$ vanishes because F_i is a fiber class, we get $\chi(Q, \mathcal{O}_Q(F_i)) = 2$ by the Riemann–Roch theorem (see [2, § I.5]). Therefore $h^1(Q, \mathcal{O}_Q(F_i)) = 0$.

For $b \in B$, let $X_b = p^{-1}(b)$ and $\mathcal{L}_{i,b} = \mathcal{L}_i|_{X_b}$. Since $h^1(Q, \mathcal{O}_Q(F_i)) = h^2(Q, \mathcal{O}_Q(F_i)) = 0$, by using the semicontinuity theorem (see [16, § 7, Satz 3]) we can and will assume, up to shrinking B , that $h^1(X_b, \mathcal{L}_{i,b}) =$

$h^2(X_b, \mathcal{L}_{i,b}) = 0$ for every $b \in B$. The class of $\mathcal{L}_{i,b}$ has self-intersection 0 by definition, so the Riemann–Roch theorem on the K3 surface X_b implies $\chi(X_b, \mathcal{L}_{i,b}) = 2$. Hence $h^0(X_b, \mathcal{L}_{i,b}) = 2$ for every $b \in B$. By Grauert’s theorem [16, § 7, Satz 5], a basis of $H^0(Q, \mathcal{O}_Q(F_i))$ extends to two sections of $\mathcal{L}_i$ whose restrictions form a basis of $H^0(X_b, \mathcal{L}_i|_{X_b})$ for every $b \in B$. As these two sections have no common zero on Q , up to shrinking B we can and will assume that they have no common zero on any fiber. Therefore, by taking ratios of the two sections we obtain morphisms $\pi_{i,b} : X_b \rightarrow \mathbb{P}^1$ which deform the three initial fibrations π_i .

The product $\Phi_b = (\pi_{1,b}, \pi_{2,b}, \pi_{3,b}) : X_b \rightarrow (\mathbb{P}^1)^3$ is an embedding for $b = 0$ by Prop. 4.2. Since being an embedding is an open property, up to shrinking again B we assert that every Φ_b is an embedding. Its image is a smooth hypersurface which has necessarily multidegree $(2, 2, 2)$ (see the argument in Prop. 4.2). Thus every X_b is Oka by Cor. 3.4.

Finally, remark that by construction Kum is primitive in the Néron–Severi group $\text{NS}(X_b)$ of every fiber X_b . By Nikulin’s criterion [26, Th. 3] we conclude that every X_b is a Kummer surface. $\square$

Proposition 4.5. *There is a marked Oka Kummer surface with period $P_{\text{irr}} \in \mathcal{D}_K$ such that $P_{\text{irr}} \cap \Lambda_{\mathbb{Q}} = \{0\}$.*

Proof. Let $B \subset \mathcal{D}_Q$ be the nonempty open subset provided by Prop. 4.4. For a nonzero vector $v \in \Lambda_{\mathbb{Q}}$, consider the locus of periods $P \in \mathcal{D}_Q$ containing v . This locus is empty unless $v \in ((\Lambda_Q)_{\mathbb{R}})^{\perp}$ and $v^2 > 0$. When it is nonempty, the positive definite oriented two-planes containing the positive line $\mathbb{R}v$ are parametrized by positive lines in $v^{\perp} \cap ((\Lambda_Q)_{\mathbb{R}})^{\perp}$. They therefore form a closed real submanifold of dimension $20 - \text{rk } \Lambda_Q$, whereas $\mathcal{D}_Q$ has real dimension $2(20 - \text{rk } \Lambda_Q)$. By (4.2) this locus is nowhere dense.

There are only countably many nonzero vectors in $\Lambda_{\mathbb{Q}}$. Since B is a Baire space, it contains a period P_{irr} which contains none of the above loci. In other words we can find a period satisfying $P_{\text{irr}} \cap \Lambda_{\mathbb{Q}} = \{0\}$. $\square$

5. PERIOD DOMAIN AND ERGODIC COMPLEX STRUCTURE

Let $O(\Lambda)$ be the isometry group of the K3 lattice Λ and let $\Gamma \subset O(\Lambda)$ be a finite-index subgroup preserving $\mathcal{D}$. By Verbitsky’s theorem [32, Th. 2.5] which is an application of Ratner’s theory [29], the lattice P_{irr} constructed in Prop. 4.5 has dense orbit in the period domain:

$$(5.1) \quad \overline{\Gamma \cdot P_{\text{irr}}} = \mathcal{D}.$$

Every point of the orbit $\Gamma \cdot P_{\text{irr}}$ is the period of a marking of the same Oka surface.

We need a similar density statement inside the Kummer period domain. Let $L = \text{Kum}^{\perp} \subset \Lambda$ and let $\Gamma_{\text{Kum}} \subset O(\Lambda)$ be the subgroup of $O(\Lambda)$ acting trivially on Kum. We assert that the image of Γ_{Kum} in $O(L)$ has finite index. Indeed, since Kum and L are primitive orthogonal complements in the unimodular lattice Λ , we have $\text{Kum} \oplus L \subset \Lambda \subset \text{Kum}^{\vee} \oplus L^{\vee}$ where $\text{Kum}^{\vee}, L^{\vee}$

denote the dual lattices. If $g \in O(L)$ acts trivially on the discriminant group $L^\vee/L$, then $g(y) - y \in L$ for every $y \in L^\vee$, so $\text{id}_{\text{Kum}} \oplus g$ preserves Λ . Thus the image of Γ_{Kum} contains $\ker(O(L) \rightarrow O(L^\vee/L))$, which has finite index.

Let Γ_{Kum}^+ be the subgroup of Γ_{Kum} preserving $\mathcal{D}_K$. Its image has finite index in the subgroup of $O(L)$ preserving $\mathcal{D}_K$. Since L has signature $(3, 3)$ and $P_{\text{irr}} \cap L_{\mathbb{Q}} = \{0\}$, Verbitsky's theorem [32, Th. 2.5] still can be applied, and we obtain

$$(5.2) \quad \overline{\Gamma_{\text{Kum}}^+ \cdot P_{\text{irr}}} = \mathcal{D}_K.$$

Every point of this orbit is again the period of a marking of the same Oka Kummer surface.

Proof of Theorem E. By (5.1), the set $\mathcal{O}$ is dense in the period domain $\mathcal{D}$. We claim that it is also G_δ . By the local Torelli theorem [19, Ch. 6,7] every point of $\mathcal{D}$ has a neighborhood biholomorphic to the base of a marked Kuranishi family. Two marked K3 surfaces with the same period have biholomorphic underlying surfaces by the global Torelli theorem [19, Ch. 7, Th. 5.3]. Thus, the intersection of $\mathcal{O}$ with such a neighborhood coincides with the Oka locus of the corresponding Kuranishi family. This locus is G_δ by Lárusson's theorem [22, Cor. 8]. Hence $\mathcal{O}$ is locally G_δ in the period domain, and therefore G_δ by Lemma 5.1 below.

For every nonzero $h \in \Lambda$, the locus $\{P \in \mathcal{D} \mid P \perp h\}$ is a closed codimension-one analytic subset. Consequently the locus $\{P \in \mathcal{D} \mid \Lambda \cap P^\perp = 0\}$ is a dense G_δ subset of $\mathcal{D}$. Its intersection with $\mathcal{O}$ is therefore again a dense G_δ subset because $\mathcal{D}$ is a Baire space. For every period in this intersection the corresponding K3 surface is nonprojective and contains no curve because its Néron–Severi group vanishes.

By (5.2), $\mathcal{O}_K$ is dense in $\mathcal{D}_K$, and it is G_δ because $\mathcal{O}_K = \mathcal{O} \cap \mathcal{D}_K$. The same argument as above shows that periods with Néron–Severi lattice exactly equal to Kum form a dense G_δ in $\mathcal{D}_K$. Indeed, if $h \in \Lambda \setminus \text{Kum}$, then its projection to $L_{\mathbb{Q}}$ is nonzero because Kum is primitive. Hence the condition $P \perp h$ defines a closed hypersurface in $\mathcal{D}_K$. Thus the Oka periods with Néron–Severi lattice exactly equal to Kum form a dense G_δ in $\mathcal{D}_K$. Since Kum is negative definite, the corresponding K3 surfaces are nonprojective by the Hodge index theorem. $\square$

Lemma 5.1. *Let M be a second-countable metrizable space and let $A \subset M$ be a subset. If A is locally G_δ , then it is G_δ .*

Proof. The metrizability ensures that every open subset of M is F_σ . Consider a countable open cover (U_j) of M such that $U_j \setminus A$ is F_σ in U_j . Every subset closed in U_j has the form $U_j \cap F$ with F closed in M , hence is F_σ in M because U_j is. Thus each $U_j \setminus A$ is F_σ in M , and so is $M \setminus A = \bigcup_j (U_j \setminus A)$. $\square$

Proof of Theorem A. Choose a marking of the local Kuranishi family and let $\text{per} : B \rightarrow \mathcal{D}$ be its period map. By the local Torelli theorem, per is a local biholomorphism.

Let $V \subset B$ be a nonempty open subset. Then $\text{per}(V)$ is open in $\mathcal{D}$. By Theorem E, it contains a period represented by an Oka K3 surface. If $b \in V$ has this period, then the corresponding K3 fiber X_b is biholomorphic to that surface by the global Torelli theorem. Therefore $V \cap B_{\text{Oka}} \neq \emptyset$. Since V was arbitrary, B_{Oka} is dense in B . It is G_δ by Lárusson's theorem [22, Cor. 8].

Similarly, Theorem E implies that every $\text{per}(V)$ contains an Oka period P satisfying $\Lambda \cap P^\perp = 0$. Since $\varphi_b(\text{NS}(X_b)) = \Lambda \cap \text{per}(b)^\perp$, the corresponding fiber X_b is Oka and satisfies $\text{NS}(X_b) = 0$. Thus these parameters are also dense in B . Moreover, the locus where $\text{NS}(X_b) = 0$ is the inverse image under per of $\{P \in \mathcal{D} \mid \Lambda \cap P^\perp = 0\}$, which is G_δ by the proof of Theorem E. Its intersection with B_{Oka} is therefore G_δ . $\square$

6. AN OKA HYPERKÄHLER FOURFOLD

An *irreducible holomorphic symplectic manifold* is a simply connected compact Kähler manifold whose space of holomorphic 2-forms is generated by a symplectic form. Such a manifold is also called an irreducible hyperkähler manifold. We say that it is of $K3^{[n]}$ -*type* if it is deformation equivalent to the Hilbert scheme of length- n subschemes of a K3 surface. For an irreducible hyperkähler manifold there is a quadratic form, called Beauville–Bogomolov–Fujiki form, on the second integral cohomology (see [3]). In this section, we only consider hyperkähler fourfolds, and denote by q the Beauville–Bogomolov–Fujiki form normalized by $\int \alpha^4 = 3q(\alpha)^2$.

6.1. Construction. Let $S \subset (\mathbb{P}^1)^3$ be a Wehler K3 surface, i.e. a smooth $(2, 2, 2)$ hypersurface. Its coordinate projections $p_i : S \rightarrow \mathbb{P}^1$, for $i = 1, 2, 3$, are genus-one fibrations. Let $F_i = c_1(p_i^* \mathcal{O}_{\mathbb{P}^1}(1))$ be their fiber cohomology classes. Let $X_0 = S^{[2]}$ be the Hilbert scheme of length-two subschemes of S ; it is a hyperkähler variety. Each p_i induces a Lagrangian fibration $f_i^0 : X_0 \rightarrow \text{Sym}^2(\mathbb{P}^1) \simeq \mathbb{P}^2$ where Sym^2 denotes the symmetric product. A general fiber of f_i^0 is a product of two genus-one curves.

Let $E \subset X_0$ be the exceptional divisor of the natural morphism $X_0 = S^{[2]} \rightarrow \text{Sym}^2(S)$, i.e. E parametrizes nonreduced length-two subschemes of S . By [3, § 6], there is an orthogonal decomposition

$$(6.1) \quad H^2(X_0, \mathbb{Z}) \simeq \Omega \oplus \mathbb{Z}\delta$$

where $2\delta = [E]$, $q(\delta) = -2$, and $q|_\Omega$ coincides with the intersection form on S . Since Ω has rank 22, the second Betti number of X_0 is $b_2(X_0) = 23$. Via the decomposition (6.1), the class $F_i \in \Omega$ corresponds exactly to the fiber class $c_1((f_i^0)^* \mathcal{O}_{\mathbb{P}^2}(1))$. In the sequel we will denote both classes by F_i .

Consider the sublattice $N := \mathbb{Z}F_1 + \mathbb{Z}F_2 + \mathbb{Z}F_3$. As $F_i^2 = 0$ and $F_i \cdot F_j = 2$ for $i \neq j$, the intersection matrix on N is

$$(6.2) \quad (q(F_i, F_j))_{i,j=1}^3 = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}, \quad \text{so } N \simeq U(2) \oplus \langle -4 \rangle,$$

where $U(2)$ has Gram matrix $\begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}$, and $\langle -4 \rangle$ has a generator of square -4 . Here the orthogonal decomposition in (6.2) is compatible to the basis $F_1, F_2, F_3 - F_1 - F_2$. In particular, N has signature $(1, 2)$. By Lefschetz Theorem the restriction $H^2((\mathbb{P}^1)^3, \mathbb{Z}/p\mathbb{Z}) \rightarrow H^2(S, \mathbb{Z}/p\mathbb{Z})$ is injective for every prime number p . Therefore N is a primitive sublattice in Ω , and hence in $H^2(X_0, \mathbb{Z})$.

By [23, § 6.2], there exist a local Kuranishi family of the pair $(X_0, (f_i^0)^* \mathcal{O}_{\mathbb{P}^2}(1))$, i.e. a smooth proper holomorphic family $\pi : \mathcal{X} \rightarrow T$ of $K3^{[2]}$ -type fourfolds over a small contractible base, with $\pi^{-1}(0) = X_0$, and line bundles $\mathcal{L}_i$ extending $(f_i^0)^* \mathcal{O}_{\mathbb{P}^2}(1)$. The fibers have markings $H^2(X_t, \mathbb{Z}) \simeq \Omega \oplus \mathbb{Z}\delta$ so that $c_1(\mathcal{L}_i|_{X_t}) = F_i$. The period map sends T biholomorphically to an open neighborhood of $H^{2,0}(X_0)$ in the period domain

$$(6.3) \quad \left\{ [w] \in \mathbb{P}(H^2(X_0, \mathbb{C})) \mid q(w) = 0, \ q(w, \bar{w}) > 0, \ q(w, F_i) = 0 \right\}.$$

As N has rank three, we have $\dim T = 23 - 3 - 2 = 18$.

Lemma 6.1. *For t sufficiently close to 0, the line bundles $\mathcal{L}_i|_{X_t}$ are globally generated and have three global sections. Their linear systems define Lagrangian fibrations onto $\mathbb{P}^2$, and the product of these three fibrations has differential of rank at least two everywhere.*

Proof. By Matsushita's theorem [24, Th. 1.2 and Cor. 1.3], the complete linear systems of $\mathcal{L}_i|_{X_t}$ define holomorphic families of Lagrangian fibrations $f_i^t : X_t \rightarrow \mathbb{P}^2$ extending f_i^0 , for t near 0. In particular $\mathcal{L}_i|_{X_t}$ is globally generated and has three global sections.

Consider a point $\xi \in X_0 \setminus E$ representing two distinct points $x, y \in S$. For u near x , we have $f_i^0(\{u, y\}) = p_i(u) + p_i(y) \in \text{Sym}^2(\mathbb{P}^1)$. Note that each map $b \mapsto b + p_i(y)$ embeds $\mathbb{P}^1$ into $\text{Sym}^2(\mathbb{P}^1)$ as a line. Since $(p_1, p_2, p_3) : S \rightarrow (\mathbb{P}^1)^3$ is an embedding, we deduce that $u \mapsto (f_1^0(\{u, y\}), f_2^0(\{u, y\}), f_3^0(\{u, y\}))$ is an immersion. Hence $d(f_1^0, f_2^0, f_3^0)_\xi$ has rank at least two.

Now we study what happens on E . Let $\rho : E \rightarrow S$ be the natural projection. For $\xi \in E$, we have $f_i^0(\xi) = 2p_i(\rho(\xi))$. As ρ is a submersion, (p_1, p_2, p_3) is an embedding, and $b \mapsto 2b$ embeds $\mathbb{P}^1$ into $\text{Sym}^2(\mathbb{P}^1)$ as a conic, we deduce that the rank of $d(f_1^0, f_2^0, f_3^0)$ on E is exactly two.

Thus the product differential has rank at least two everywhere on X_0 . The same holds for $d(f_1^t, f_2^t, f_3^t)$ for t sufficiently close to 0. $\square$

For every class $\alpha \in \Omega \setminus N$, the equation $q(w, \alpha) = 0$ defines a proper closed analytic subset of T . We may choose t arbitrarily close to 0, satisfying Lemma 6.1 and avoiding all these countably many subsets. Let $X := X_t$. Then

$$(6.4) \quad \text{NS}(X) = N.$$

Let $f_i : X \rightarrow \mathbb{P}^2$ be the Lagrangian fibrations defined by the complete linear systems of $\mathcal{L}_i|_X$, and fix a holomorphic symplectic form σ on X .

Lemma 6.2. *The hyperkähler fourfold X is projective. Its positive cone is its Kähler cone. The class $F_1 + F_2$ is ample while $(f_1, f_2) : X \rightarrow (\mathbb{P}^2)^2$ is a finite morphism.*

Proof. The positive cone of X is the component of $\{\beta \in H^{1,1}(X, \mathbb{R}) \mid q(\beta) > 0\}$ containing the Kähler classes. As every element of N has a square divisible by four, there is no integral $(1, 1)$ -class on X of square -2 or -10 by (6.4). Then by [25, Prop. 2.12], we deduce that the Kähler cone equals the positive cone.

The class $F_1 + F_2$ belongs to the positive cone and is a Kähler class. By Kodaira's embedding theorem, it is ample and X is projective. Since a positive-dimensional fiber of (f_1, f_2) would contain a curve on which the corresponding ample line bundle has degree zero, the map $(f_1, f_2) : X \rightarrow \mathbb{P}^2 \times \mathbb{P}^2$ is finite. $\square$

6.2. Automorphisms. Let $N_{\mathbb{R}} := N \otimes_{\mathbb{Z}} \mathbb{R}$, and let $O^+(N)$ be the group of integral isometries preserving the component of $\{\beta \in N_{\mathbb{R}} \mid q(\beta) > 0\}$ containing $F_1 + F_2$. Denote by $N^{\vee}$ the dual lattice $\{u \in N \otimes_{\mathbb{Z}} \mathbb{Q} \mid q(u, N) \subset \mathbb{Z}\}$. The discriminant group $N^{\vee}/N$ is finite, equipped with a quadratic form $[u] \mapsto q(u) \bmod 2\mathbb{Z}$. We denote its isometry group by $O(N^{\vee}/N)$, and the real algebraic group of determinant-one isometries of $N_{\mathbb{R}}$ by $SO(N_{\mathbb{R}}, q)$.

Let $N^{\perp}$ be the orthogonal complement of N in $H^2(X, \mathbb{Z})$, and let $G := \{g \in \text{Aut}(X) \mid g^*|_{N^{\perp}} = \text{id}\}$. As the symplectic class satisfies $\sigma \in N^{\perp} \otimes \mathbb{C}$, every element of G preserves σ .

Lemma 6.3. *The image of G in $O(N)$ contains the kernel of $O^+(N) \rightarrow O(N^{\vee}/N)$. Every finite-index subgroup of G acts irreducibly on $N_{\mathbb{R}}$.*

Proof. Let $\gamma \in \ker(O^+(N) \rightarrow O(N^{\vee}/N))$. Extend γ to a rational isometry $\tilde{\gamma}$ of $H^2(X, \mathbb{Q})$ by using the identity on $N^{\perp} \otimes \mathbb{Q}$. For $\alpha \in \Omega$, we denote by α_N its orthogonal projection to $N \otimes \mathbb{Q}$. Then $\alpha_N \in N^{\vee}$, so $\tilde{\gamma}(\alpha) - \alpha = \gamma(\alpha_N) - \alpha_N \in N$. Hence $\tilde{\gamma}(\Omega) = \Omega$. Since $\tilde{\gamma}$ fixes the class δ (defined in (6.1)), it defines an integral isometry of $H^2(X, \mathbb{Z})$.

Note that $\tilde{\gamma}$ fixes σ and preserves the positive cone. Since Ω is unimodular and $q(\delta) = -2$, the discriminant group of $H^2(X, \mathbb{Z})$ is $\mathbb{Z}/2\mathbb{Z}$, generated by the class of $\delta/2$. We remark that $\tilde{\gamma}$ acts trivially on the discriminant group of $H^2(X, \mathbb{Z})$. By [23, Def. 1.1 and Lem. 9.2], $\tilde{\gamma}$ is a parallel-transport operator. Recall that by Lem. 6.2 the Kähler cone equals the positive cone. Therefore we can apply the Global Torelli Theorem [23, Th. 1.3] to find $g \in G$ such that $g^* = \tilde{\gamma}$.

Recall that $F_1, F_2, (F_3 - F_1 - F_2)$ is a basis of N . We will think of an element $xF_1 + yF_2 + z(F_3 - F_1 - F_2) \in N$ as a symmetric matrix $M = \begin{pmatrix} x & z \\ z & y \end{pmatrix}$. Then $q(M) = 4\det M$, and $F_1 + F_2$ corresponds to the identity matrix. Hence the action $M \mapsto AMA^t$ defines a homomorphism $\text{SL}_2(\mathbb{Z}) \rightarrow O^+(N)$.

Let $H \subset G$ be a subgroup of finite index. Then the image of H has finite index in $O^+(N)$. Its inverse image $\Gamma \subset \text{SL}_2(\mathbb{Z})$ therefore has finite index and

is Zariski-dense in $\mathrm{SL}_2(\mathbb{R})$. Since the representation of $\mathrm{SL}_2(\mathbb{R})$ on symmetric matrices is irreducible, we conclude that H acts irreducibly on $N_{\mathbb{R}}$. $\square$

6.3. Invariant surfaces. For an irreducible subvariety $Z \subset X$, we denote by Z_{reg} its smooth locus, and define $\mathrm{Stab}_G(Z) := \{g \in G \mid g(Z) = Z\}$.

Lemma 6.4. *If an irreducible surface $Z \subset X$ has finite orbit under G , then $\sigma|_{Z_{\mathrm{reg}}} = 0$.*

Proof. Let $\mu : \tilde{Z} \rightarrow X$ be the minimal resolution of singularities (see [2, Th. III.6.2]). The action of $\mathrm{Stab}_G(Z)$ lifts to $\tilde{Z}$ by uniqueness of the resolution. Note that $\mathrm{Stab}_G(Z)$ has finite index in G because Z has finite orbit.

The map $\beta \mapsto \int_Z \beta^2$ defines a quadratic form on $N_{\mathbb{R}}$ which is invariant under $\mathrm{Stab}_G(Z)$. By representation-irreducibility of $\mathrm{Stab}_G(Z)$ from Lemma 6.3, there exists $c > 0$ such that

$$(6.5) \quad \int_{\tilde{Z}} \mu^* \beta \mu^* \gamma = \int_Z \beta \gamma = c q(\beta, \gamma), \quad \beta, \gamma \in N_{\mathbb{R}}.$$

Suppose now for contradiction that $\mu^* \sigma \neq 0$. Then the kernel of $\mu^* : H^2(X, \mathbb{Q}) \rightarrow H^2(\tilde{Z}, \mathbb{Q})$ is a rational Hodge substructure without $(2, 0)$ -part or $(0, 2)$ -part because $H^{2,0}(X) = \mathbb{C}\sigma$. By the Lefschetz $(1, 1)$ -theorem, it is contained in $\mathrm{NS}(X) \otimes \mathbb{Q} = N \otimes \mathbb{Q}$. If β is in this kernel, then by (6.5) $q(\beta, \gamma) = 0$ for every $\gamma \in N \otimes \mathbb{Q}$. As q is nondegenerate on N , we have $\beta = 0$. Thus μ^* is injective.

Let R be the zero divisor of $\mu^* \sigma$. Since G preserves σ , the divisor R is invariant under $\mathrm{Stab}_G(Z)$. By Lemma 6.3, the invariant linear functional $\beta \mapsto R \cdot \mu^* \beta$ vanishes on $N_{\mathbb{R}}$. For an ample class β , the degree of $\mu^* \beta$ on a curve is positive unless the curve is contracted by μ . Therefore every component of R is contracted by μ .

For every $m \geq 1$ and nonzero $s \in H^0(\tilde{Z}, mK_{\tilde{Z}})$, the poles of $s/(\mu^* \sigma)^m$ are supported on R . Being a rational function on $\tilde{Z}^\nu$, $s/(\mu^* \sigma)^m$ has no poles in codimension one. It is thus constant. Therefore $h^0(\tilde{Z}, mK_{\tilde{Z}}) = 1$ for every $m \geq 1$. By the classification of projective surfaces [2], there is a birational morphism $\rho : \tilde{Z} \rightarrow Z_0$ where Z_0 is K3 or abelian. Hence the form $\mu^* \sigma$ descends to a nowhere-vanishing holomorphic 2-form on Z_0 . This implies that every ρ -exceptional curve occurs in R , thus is contracted by μ . We then obtain

$$(6.6) \quad \mu^* H^2(X, \mathbb{Q}) \subset \rho^* H^2(Z_0, \mathbb{Q}).$$

By the injectivity of μ^* and (6.6), we deduce $23 = b_2(X) \leq b_2(Z_0) \leq 22$, a contradiction. $\square$

6.4. The Oka property. We will use the terminology of Section 3. By Prop. 3.2, each f_i induces an additive spray $s_i : f_i^* T_{\mathbb{P}^2}^* \rightarrow X$ with vertical

differential $A_i = \sigma^\sharp \circ df_i^*$. Let $\mathcal{P}$ be the pseudogroup generated by G and the local biholomorphisms induced these three sprays, as in § 3.4. Define

$$(6.7) \quad D := \left\{ x \in X \mid \bigcap_{i=1}^3 \ker(df_{i,x}) \neq \{0\} \right\}.$$

Theorem 6.5. *The projective hyperkähler fourfold X is Oka.*

Proof. The set D is algebraic. Since (f_1, f_2) is finite by Lem. 6.2, its differential has rank four on a nonempty Zariski-open subset, so $D \neq X$. The locus defined in (3.5) is $U = X \setminus D$.

Suppose now for contradiction that a $\mathcal{P}$ -orbit $\mathcal{O}$ is contained in D . Let $W := \overline{\mathcal{O}}^{\text{Zar}}$. Then $W \subset D$ is a G -invariant algebraic subset.

Let us fix $i \in \{1, 2, 3\}$, a Zariski-open set $V \subset \mathbb{P}^2$, and an algebraic one-form a on V . On $f_i^{-1}(V)$, recall that in Section 3 we defined the vector field $V_{i,a}(x) = A_{i,x}(a(f_i(x)))$. As σ and f_i are algebraic, $V_{i,a}$ is algebraic. For a regular local function φ vanishing on W and a point $x \in \mathcal{O}$ where φ is defined, we have

$$V_{i,a}(\varphi)(x) = \left. \frac{d}{dt} \right|_{t=0} \varphi(s_i(x, t a(f_i(x)))) = 0$$

because the local biholomorphisms in $\mathcal{P}$ preserve $\mathcal{O}$. By Zariski density, the local regular function $V_{i,a}(\varphi)$ vanishes on W wherever it is defined. Let Z be an irreducible component of W . Then $V_{i,a}$ is tangent to Z at its smooth points outside the intersection locus with other components of W . The above arguments show that, at a general point $x \in Z_{\text{reg}}$, we have

$$(6.8) \quad A_{i,x}(T_{f_i(x)}^* \mathbb{P}^2) \subseteq T_x Z, \quad i = 1, 2, 3.$$

If $\dim Z = 0$, then $A_{i,x} = 0$ for every i by (6.8). By (3.3), we would have $df_{i,x} = 0$ for every i , contradiction to Lemma 6.1.

If $\dim Z = 3$, then $[Z] \in N$ would be fixed by $\text{Stab}_G(Z)$ which is a finite-index subgroup of G because G permutes the components of W . This would contradict Lem. 6.3.

If $\dim Z = 1$, then $\beta \mapsto \int_Z \beta$ would be a nonzero linear functional invariant under $\text{Stab}_G(Z)$, again a contradiction to Lem. 6.3.

Therefore $\dim Z = 2$, and by Lemma 6.4 we have $\sigma|_{Z_{\text{reg}}} = 0$. For $i = 1, 2$, $v \in T_x Z$ and $\lambda \in T_{f_i(x)}^* \mathbb{P}^2$, by (3.3) and (6.8) we have

$$\lambda(df_{i,x}(v)) = \sigma_x(A_{i,x}(\lambda), v) = 0.$$

Hence both f_1 and f_2 are constant on Z . This contradicts the finiteness of (f_1, f_2) proved in Lem. 6.2.

Every $\mathcal{P}$ -orbit therefore meets U , and we conclude by Prop. 3.6. $\square$

As our fourfold X was constructed by choosing a Baire generic parameter t (see the end of § 6.1), we proved:

Corollary 6.6. *There are infinitely many Oka projective hyperkähler manifolds.*

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AI USAGE

The authors developed the core materials on K3 surfaces for Th. A. The hyperkähler example is found through interactions of the authors with LLM models. LLM models have also been used for the editorial work of the manuscript. The authors take full responsibility of the paper.

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